

Plano principal de APP

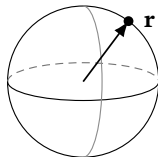
Representación de Pauli

Esfera de Bloch

$$\begin{aligned} S^\top &= S, & J^\top &= -J \\ S^2 &= I, & J^2 &= -I \\ SJ &= -JS \end{aligned}$$



$$\begin{aligned} \sigma_1 &= SJ \\ \sigma_2 &= -iJ \\ \sigma_3 &= S \end{aligned}$$



complejificación

estados puros

$$\rho_\psi = \psi\psi^\dagger = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$$

$$h : r_k = \psi^\dagger \sigma_k \psi$$

$$S^1 \quad \xrightarrow{\quad \quad \quad}$$

$$S^3 \quad \xrightarrow{\quad \quad \quad}$$

$$S^2 \simeq \mathbb{CP}^1$$

fase común

vector unitario

rayo

$$e^{i\chi}\psi$$

$$\psi^\dagger\psi = 1$$

$$\|\mathbf{r}\| = 1$$